

LogProof - Ensuring Integrity and Immutability of Audit Logs Using Blockchain

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Abstract—Ensuring the integrity and immutability of audit logs is critical for preserving trust, enabling forensic accuracy, and meeting stringent regulatory requirements in systems handling sensitive data. Conventional centralized logging approaches, while widely used, remain vulnerable to insider manipulation, unauthorized modification, and single points of failure posing substantial security and compliance risks. This paper presents LogProof, a blockchain based software framework that delivers a decentralized, tamper-evident solution for secure audit log management. LogProof integrates seamlessly with existing logging infrastructures, computing cryptographic hashes of log entries and immutably recording them on a blockchain, with each entry linked to a unique transaction ID for independent verification. This approach eliminates reliance on trusted third parties, enhances accountability through transparent verification, and supports scalability via off-chain log storage. The framework is validated through a containerized prototype using Multi-Chain, demonstrating real-time tamper detection and efficient verification. LogProof provides a practical, adaptable foundation for secure audit logging, with applicability across sectors such as finance, healthcare, and government, and offers a platform for advancing blockchain adoption in data integrity critical environments.

Index Terms—audit logs, blockchain, compliance, immutable, integrity, secure logging frameworks, tamper detection

I. INTRODUCTION

In today's digital landscape, the integrity and immutability of audit logs are critical for ensuring trust, accountability, and compliance in systems that handle sensitive data. Audit logs serve as an immutable record of events, detailing the activities and actions performed within a system. They play a pivotal role in detecting security incidents, preventing fraud, and maintaining compliance with industry regulations such as GDPR, HIPAA, and SOX [1]–[3]. However, audit logs are susceptible to tampering, deletion, or unauthorized modifications, posing a significant risk to the integrity of systems and organizations [4], [5]. Secure access control mechanisms are also critical to protect audit logs from unauthorized modifications [6].

Ensuring the trustworthiness of these logs has traditionally been challenging. Conventional methods such as centralized logging systems, file integrity monitoring, and cryptographic

signing offer some level of protection but often fall short in guaranteeing the immutability and integrity of audit records, especially in environments where logs can be altered by privileged users or malicious insiders [7]. These traditional systems depend on a trusted central authority, which itself could become a single point of failure or compromise [8], [9].

II. MOTIVATION

Blockchain presents a decentralized and secure method for ensuring the integrity of data. Initially developed for cryptocurrencies like Bitcoin, blockchain has since evolved into a versatile tool for any application requiring tamper-proof records [10], [11]. At its core, blockchain is a chain of blocks, each containing a cryptographically linked list of transactions. These blocks are appended in a linear, immutable sequence, making it computationally infeasible to alter any previous record without detection. This property is crucial for audit logs, as it guarantees the immutability and transparency of the recorded events [12].

The use of blockchain for audit logging provides several advantages over traditional approaches:

- **Decentralization:** Blockchain eliminates the need for a central authority or trusted third party, ensuring that no single entity has control over the audit logs. This reduces the risk of tampering by insiders or privileged users.
- **Immutability:** Once data is written to the blockchain, it cannot be modified without the consensus of the entire network, ensuring that logs are tamper-resistant. This makes blockchain particularly suitable for environments where logs are highly sensitive and must be preserved in their original state [13].
- **Transparency and Accountability:** Blockchain's transparent nature allows stakeholders to independently verify the integrity of audit logs. This is especially important in industries where compliance and accountability are paramount [14], [15].
- **Scalability with Off-Chain Solutions:** By storing the log data off-chain and using the blockchain to store

cryptographic references (hashes), it is possible to achieve scalability without compromising the integrity or immutability of the logs [16].

III. PROPOSED DESIGN - LOGPROOF FRAMEWORK

The LogProof framework is designed to ensure the immutability and integrity of audit logs by leveraging blockchain technology. The following sections outline the integration of LogProof within general software systems, enabling tamper-proof log verification through cryptographic hashing and blockchain storage.

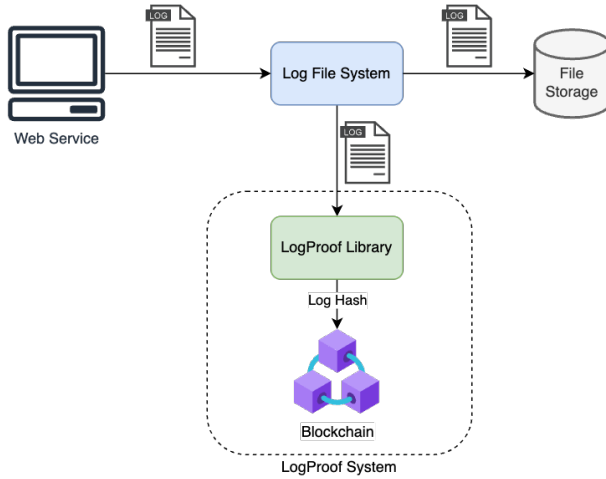


Fig. 1. Storing log hash in LogProof system.

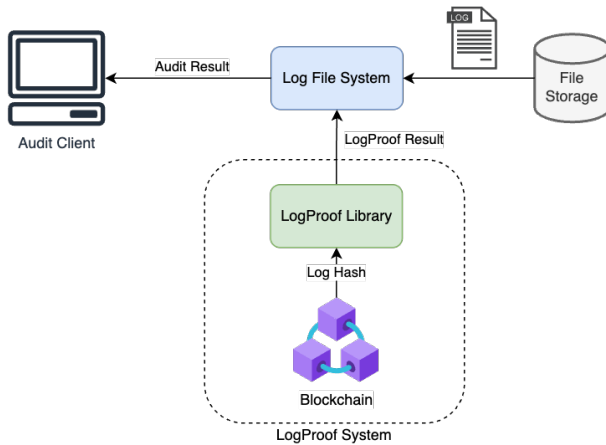


Fig. 2. Verifying log hash in LogProof system.

A. Integration with Logging Libraries

In a typical software system, log data is generated and stored using a logging library. The LogProof system is designed to integrate seamlessly with these existing logging mechanisms.

The LogProof module provides a library that allows applications to interact with a blockchain, enabling them to publish log hashes and retrieve verification data from the blockchain [17]. This integration ensures that no significant changes are required to the existing logging infrastructure, allowing for easy adoption.

B. Storing Log Hash on Blockchain

When an application generates a log, the LogProof system intercepts the log data and calculates its cryptographic hash. This hash is then submitted to the blockchain via the LogProof module, where it is stored immutably, as shown in Fig. 1. In response, the blockchain returns a transactionId, which serves as a unique identifier for the logged data on the blockchain. After this process, the original log data continues to be stored as usual in the application's existing log storage system (e.g., file system, database). The only addition is that the transactionId from the blockchain can be recorded alongside the log data for future verification.

C. Verification of Log Integrity

To verify the integrity of stored logs, any audit system or administrator can use the LogProof library to retrieve the corresponding hash from the blockchain by referencing the transactionId, as shown in Fig. 2. The log data stored off-chain is hashed again, and this new hash is compared to the original hash stored on the blockchain. If the hashes match, the log data remains untampered, ensuring its integrity. However, if a mismatch occurs, it indicates that the log has been altered, triggering an alert or corrective action based on the application's security policies. Efficient blockchain querying enhances the performance of this verification process [18].

IV. EXPERIMENTAL RESULTS

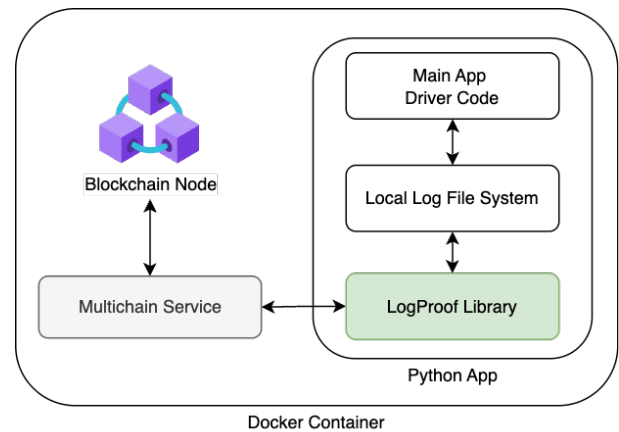


Fig. 3. Docker Container Components.

A. Container Setup

The software repository contains a docker compose configuration which deploys the docker container shown in Fig. 3. The multichain service sets up a blockchain node, initializes a stream, and subscribes to it. The python app service runs a Python application that interacts with the MultiChain node over its RPC API. The Python driver code connects to the MultiChain node, writes audit logs to the local file system, and verifies the integrity of each log by comparing it with the corresponding blockchain hash.

B. Container Output

The output of the Docker container is shown in Code Snippet 1. We first verified the hash of the stored log against the hash on the blockchain, and then simulated tampering with the log, which was correctly detected by the framework.

```
multichain-node | MultiChain 2.3.2 Utilities (latest
↳ protocol 20013)
...
multichain-node | MultiChain 2.3.2 Daemon (Community Edition
↳ , latest protocol 20013)
multichain-node | Starting up node...
...
multichain-node | Multichain node is ready.
...
python-app      | Connected to MultiChain.
...
python-app      | Stream 'stream1' is available.
...
python-app      | Attempting to store log hash on blockchain
python-app      | Log hash stored on blockchain. txid:
python-app      | 3
↳ a13774285576adbeb6a78f5315eff05fe7d068d74ecf3b000dd47c6
↳
↳ d0fd5c2a
python-app      | Log written to /app/logs/audit.log
python-app      | Content of audit.log:
python-app      | ['3
↳ a13774285576adbeb6a78f5315eff05fe7d068d74ecf3b000dd47c6
↳
↳ d0fd5c2a
python-app      | -2024-10-13 04:59:41-AUDIT-User login
↳ event at 10:01 AM']
python-app      | Log verified: No tampering detected.
python-app      | Log tampered!
```

Listing 1. Code Output of running the Docker container

V. CONCLUSION

The proposed LogProof framework demonstrates a practical, scalable, and tamper-proof approach to safeguarding the integrity of audit logs through blockchain technology. By leveraging cryptographic hashing and decentralized ledger storage, LogProof effectively mitigates the risks inherent in traditional centralized logging systems, particularly vulnerabilities to insider tampering and single points of failure. The prototype implementation using MultiChain and Docker validates the framework's ability to detect unauthorized log modifications in real time, while its seamless integration with existing logging libraries lowers the barrier to adoption in enterprise environments.

Beyond technical feasibility, LogProof holds significant potential for industries where trust, transparency, and regulatory compliance are paramount such as finance, healthcare, and government by providing verifiable, immutable records that

can withstand both internal and external scrutiny. The off-chain storage strategy ensures scalability, enabling the solution to handle high log volumes without compromising performance or integrity.

Future work will focus on expanding platform compatibility, integrating advanced blockchain querying optimizations, conducting large-scale performance benchmarking, and aligning the framework with specific regulatory mandates such as HIPAA, GDPR, and SOX through feature-to-requirement mapping. Additionally, incorporating privacy-preserving techniques, such as zero-knowledge proofs, and AI-driven anomaly detection can further strengthen its security posture and operational value.

In essence, LogProof establishes a foundation for the next generation of secure audit log management systems, offering a compelling blueprint for organizations seeking immutable, transparent, and verifiable logging infrastructures in an increasingly security-conscious digital world.

VI. CODE AVAILABILITY

To support reproducibility and encourage further research, the complete source code for the *LogProof* framework is publicly available on GitHub at:

https://github.com/akshaynagpal/immutable_audit_logs

The repository contains:

- **Docker Compose Configuration** – for deploying a MultiChain blockchain node and the Python application.
- **LogProof Python Module** – implementing the integration with existing logging systems, log hashing, blockchain publishing, and verification routines.
- **Sample Logging Scripts** – demonstrating how to generate logs, store hashes on the blockchain, and perform verification checks.
- **Setup Instructions** – step-by-step guidance for installation, environment configuration, and running the demonstration.
- **Demonstration Scenarios** – including both normal operation and intentional log tampering to validate detection capabilities.

Researchers, developers, and security practitioners can clone the repository, deploy the containerized environment, and adapt the code to their specific audit logging requirements. Contributions via pull requests are welcome, enabling the community to extend the framework with new blockchain integrations, enhanced verification techniques, and compliance-focused features.

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